



Software Requirements Specification



Nestora

Smart Real Estate Management Platform

Revision History

Version	Author	Description	Date
1	Nestora Team	Smart Real Estate Management Platform	27/10/2025

Approval history

Approver	Role	Version	Status	Date
@nurin	Client	1	 Approved	28/10/2025

1. Introduction

1.1 Purpose

The purpose of this SRS document is to define the functional and non-functional requirements for the Nestora Real Estate Management Platform, which helps buyers, sellers, tenants, property owners, and real estate agents manage property searching, buying, selling, renting, and property-related services through a secure and intelligent digital platform.

1.2 Scope

Nestora is a smart real estate management platform that connects property buyers, sellers, tenants, property owners, and verified real estate agents across Bangladesh. The platform provides a centralized marketplace where users can browse verified residential, commercial, rental, and land properties, schedule property visits, compare listings, communicate with property owners, and manage all property-related activities from one platform.

1.3 Objectives

Nestora is a smart real estate management platform that connects buyers, sellers, tenants, property owners, and verified real estate agents across Bangladesh.

Build a robust, scalable real estate marketplace entirely in Kotlin.

Ensure low-latency REST APIs, secure property booking, user authentication, and strong data protection.

Deliver a seamless multi-platform experience via Kotlin Multiplatform (Android, iOS, Web) with real-time notifications and location-based property services.

Create and manage property listings (Residential, Commercial, Rental, and Land).

Manage property bookings, listing details, and customer interactions.

Users Can	Property Owners / Agents Can
• Register and manage multiple property preferences. • Search residential, commercial, rental, and land properties. • Book property visits and receive reminder notifications. • Save favorite properties to Wishlist. • Compare property listings. • Track booking history and inquiries. • Contact property owners and agents. • Buy, sell, or rent properties securely.	• Create and manage property listings. • Manage property bookings, listing details, and customer interactions. • Update property information and availability status. • Respond to customer inquiries. • Monitor listing performance and booking history.

1.4 Overview

The system will be built using:

- Backend: Kotlin + Spring Boot + MySQL
- Frontend: Kotlin Multiplatform / Compose Multiplatform (Web + Mobile)
- Architecture: Service-oriented microservices
- Deployment: Docker, Kubernetes, and CI/CD integration

2. Overall description

2.1 Product Perspective

Nestora is a modular Kotlin-based microservice platform, where core features like user management, property listings, booking, property management, and notifications operate as independent services integrated via secure REST APIs.

Key Integrations

Payment Gateways: PayPal, Stripe, SSLCommerz

External APIs: Google Maps API, Location API, Currency Exchange API

Notifications: Firebase Cloud Messaging (Push), Twilio (SMS)

Architecture Layers

Presentation Layer: Kotlin Compose Multiplatform UI

Business Layer: Spring Boot (Kotlin) Services

Data Layer: MySQL, Redis, Elasticsearch

Integration Layer: External APIs & Payment Gateways

2.2 User Classes and Characteristics

User Class	Description	Core Actions
Buyer	End-user searching for properties	Search properties, compare listings, save wishlist, book property visits, contact owners
Property Owner	Property owner listing properties	Create/manage property listings, update property details, manage bookings, interact with customers
Real Estate Agent	Professional property service provider	Manage multiple property listings, respond to inquiries, handle bookings, update property status
Admin	Platform monitoring and operations	Verify property listings, manage users, handle reports, monitor platform activities

2.3 Operating Environment

Backend: Kotlin 1.9+ with Spring Boot 3.x

Frontend: Kotlin Multiplatform / Compose Multiplatform (Android, iOS, Web)

Database: MySQL 8+

Cache & Search: Redis or Elasticsearch

OS & Deployment: Docker containers orchestrated via Kubernetes

Authentication & Security: OAuth2 / JWT-based authentication

Build Tools: Gradle Kotlin DSL

Testing: Ktor Test, MockK, and JUnit5

3. Functional Requirements

3.1 User Management

Register/Login using email, phone number, or OTP.
Role-based access (Buyer, Property Owner, Real Estate Agent, Admin).
Profile creation including personal and contact information.
Secure authentication using JWT and encrypted data handling.
Update account settings, password, and profile picture.
Property Owner / Agent verification (NID / Trade License / KYC upload).

3.2 Property Search & Discovery

Search residential, commercial, rental, and land properties by location.
Filter results by price range, property type, number of bedrooms, area size, availability, and amenities.
Nearby properties displayed using map integration (GPS).
View property details including pricing, location, owner information, images, and available facilities.

3.3 Property Listing & Management

Property Owners and Agents can create, edit, update, or remove property listings.
Upload multiple property images, descriptions, pricing, and location details.
Manage property availability (Available, Sold, or Rented).
View listing performance and customer inquiries.

3.4 Property Visit Booking

Create, view, reschedule, or cancel property visit appointments.

Property Owners and Agents manage availability and confirm/cancel bookings.

Real-time booking status updates and notifications.

Booking history available for both users and property owners.

3.5 Reviews & Ratings

Buyers and Tenants can rate properties and property owners.

1–5 rating scale with text reviews and image upload for property visit proof.

Admin moderation to filter fake/spam reviews.

3.6 Property Owner / Agent Dashboard

Property Owners and Agents can manage property listings and bookings.

View inquiries, booking history, and customer analytics.

Update property information, availability, and pricing.

Download property performance reports.

3.7 Admin Panel

Manage all users, properties, owners, and agents.

Verify property listings and owner/agent registrations (KYC).

Monitor platform activities and review reports.

Handle user complaints, disputes, and fraudulent listings.

Generate system-wide monthly analytics reports.

3.8 Notification System

Push, email, and SMS notifications.

Booking reminders for scheduled property visits.

Property availability alerts and status updates.

Admin broadcast messages for new property launches, offers, and announcements.

4. Non-Functional Requirements

4.1 Performance Requirements

90% of app actions and API responses complete within < 300 ms.
Instant push notifications for booking confirmations and property updates.
Fast property search and filtering with optimized database queries.

4.2 Security Requirements

JWT-based authentication and secure login.
Encrypted user and property data (HTTPS).
Property Owner and Real Estate Agent verification for secure property listings.
GDPR-style data privacy compliance for user and property information.

4.3 Scalability Requirements

Supports growing number of buyers, sellers, tenants, property owners, and agents.
Horizontal scaling using microservices for property listings and booking modules.

4.4 Reliability

99.9% uptime with failover backup servers.
Real-time notification reliability for booking confirmations and property updates.

4.5 Maintainability

Modular Kotlin microservices architecture.

Easy to update property listing features, booking modules, and user management services.

4.6 Usability

User-friendly UI for all age groups.

Easy navigation for searching, buying, selling, renting, and booking property visits.

Multi-language support planned (English & Bangla).

4.7 Availability

Cloud-native architecture with load balancing to ensure real-time property search, booking, and listing management.

4.8 Portability

Kotlin Multiplatform support for Android, iOS, and Web, allowing users to access real estate services across all devices.

4.9 Backup

Daily incremental backups + weekly full database backups to protect property listings, booking history, and user information.

5. System Design Overview

5.1 Architecture

Type: Kotlin-based Microservice Architecture
Microservices include user management, property listings, booking, notifications, property management, and admin management modules.

5.2 Communication

REST APIs (JSON) using Spring WebFlux + Kotlin Coroutines
Used for user authentication, property search, booking services, notifications, and property management.

Service	Description
User Service	Handles user authentication, buyer profiles, seller profiles, property owner accounts, and user management.
Property Service	Manages property listings, property details, images, categories, availability, and property verification.
Booking Service	Handles booking lifecycle for property visits, confirmations, cancellations, and booking history.
Notification Service	Manages push notifications, SMS, email alerts, booking reminders, and property updates.
Admin Service	Handles property verification, user management, reports, complaints, fraud detection, and platform monitoring.

5.3 Database

MySQL (JPA / Exposed ORM)

Stores user profiles, property listings, booking information, property details, wishlist, reviews, and platform data.

5.4 Cache

Redis

Used for caching property search results, booking availability, notifications, and session data.

5.5 Search

Elasticsearch

Location-based search for residential, commercial, rental, and land properties with advanced filtering.

5.6 Logging & Monitoring

Logback + Micrometer + Grafana

Tracks API calls, property listings, booking activities, user actions, and overall system health.

6. User Interface Overview

6.1 Primary Screens

Screen	Description
Home Page	Real estate dashboard with quick access to Property Search, Buy, Rent, Sell, Featured Properties, search bar, and latest property listings.
User Dashboard	Property profiles, booking history, saved properties, inquiries, wishlist, and account management.
Property Listings Page	Browse verified residential, commercial, rental, and land properties with advanced search and filtering options.
Search Results	Displays filtered property listings based on user preferences such as location, price, property type, bedrooms, and area size.
Property Details Page	Property images, pricing, location, owner information, amenities, availability, property description, reviews, and booking options.
Property Owner / Agent Dashboard	Manage property listings, booking requests, customer inquiries, property availability, and performance analytics.
Property Visit Booking Flow	Property Details → Select Date & Time → Confirm Booking → Owner Approval → Booking Confirmation
Wishlist Page	View and manage saved properties for future comparison and purchase decisions.
Admin Console	Verify property listings, manage users, monitor reports, resolve complaints, and view system analytics.

6.2 Navigation Flow

Property Search Flow

Home → Search Properties → Search Results → Property Details → Schedule Property Visit → Booking Confirmation → User Dashboard

Property Listing Flow

Home → Property Owner Dashboard → Create Property Listing → Upload Property Details → Submit for Verification → Listing Published

Booking Flow

Home → Property Details → Select Visit Date & Time → Confirm Booking → Booking History

Wishlist Flow

Home → Search Results → Property Details → Add to Wishlist → User Dashboard

Admin Flow

Admin Login → Dashboard → Verify Property Listings → Manage Users → Review Reports → Generate Analytics

Future Enhancements

- AI Property Recommendation System
- Property Price Prediction
- Google Maps Integration
- Virtual Property Tours (360° View)
- Secure Online Payment Gateway
- Mortgage & EMI Calculator
- Market Price Analytics
- Mobile Application (Android & iOS)
- AI Chatbot for Customer Support
- Real-Time Notifications & Messaging
- Property Comparison Tool
- Saved Search & Smart Alerts
- Document Upload & Digital Verification
- E-Agreement & Digital Signature
- Property Verification & KYC System
- Multi-Language Support (English & Bangla)

Conclusion

Nestora bridges the gap between property owners, buyers, tenants, and real estate professionals through a secure and intelligent digital platform. By combining modern technology with an intuitive user experience, Nestora transforms property management into a fast, efficient, and transparent process while providing a strong foundation for future academic projects, industry applications, and real-world business growth.